

Experimental Data-driven Forward and Inverse SOA WDM Gain and SNR Modelling using Gaussian Process Regression

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Abstract We experimentally demonstrate data-driven Gaussian process regression for SOA modelling in a worst-case WDM system, with 0.1dB gain and 0.23dB SNR mean absolute error. Inverse modelling is demonstrated for estimating the required SOA injection current, temperature, and input channel powers to achieve channel-specific SNR targets. ©2026 The Author(s)

Introduction

Semiconductor optical amplifiers (SOAs) are promising alternatives to doped-fibre amplifiers, combining wideband amplification, monolithic integration with transceivers, low device cost, and low power consumption. However, SOAs exhibit strong nonlinearities arising from carrier depletion, gain saturation, and self-phase modulation, thereby introducing signal-dependent nonlinear distortions [1]. These nonlinear effects can significantly degrade signal quality at high injection currents and input powers. The underlying mechanisms of SOA-induced distortions have been extensively studied, and physical device parameters, including the unsaturated gain, linewidth enhancement factor, carrier lifetime, and saturation power are known to describe the nonlinear noise.

Recently, SOAs have attracted significant attention and have been applied in various scenarios, including pre-amplification in passive optical networks [2], data-centre C+L band inter-connects [3], and inline amplification for wavelength division multiplexing (WDM) transmission systems [4]. With their capability of providing simultaneous amplification over multiple bands, SOAs are promising candidates for future ultra-wideband (UWB) systems. More recently, a GN model [5] was proposed and an SOA reservoir model [6] was experimentally verified. Meanwhile, data-driven approaches for

modelling optical amplifiers utilising machine/deep learning algorithms have gained increasing interest. Related research includes inverse Raman Amplifier design with DNN [7], EDFA gain estimation with physics-informed Neural Networks [8], and an ANN-based SOA model [9] based on the simulated Connely model [10]. However, to the best of our knowledge, few studies have demonstrated data-driven approaches to modelling SOA WDM gain and SNR using experimentally measured data. To further bridge this research gap, we propose a data-driven uncertainty-aware SOA modelling framework for 16 C-band WDM channels using Gaussian process regression (GPR). Beyond forward prediction, we extend this framework to inverse modelling, enabling reliable estimation of configuration parameters required for achieving specific SNR targets, which is a challenging task due to the lack of an analytically solvable inverse for SOA systems.

Experimental Setup and Data Collection

Our SOA WDM experimental setup is shown in Fig. 1(a), which consists of two measurement groups, namely gain and SNR. The transmitted signal is generated through the use of 16 integrated tunable laser assemblies (iTLAs), which are then bulk-modulated using a modified Ciena WaveLogic 3 (WL3) line card. It is worth noting that we deliberately demonstrate a worst-case WDM system

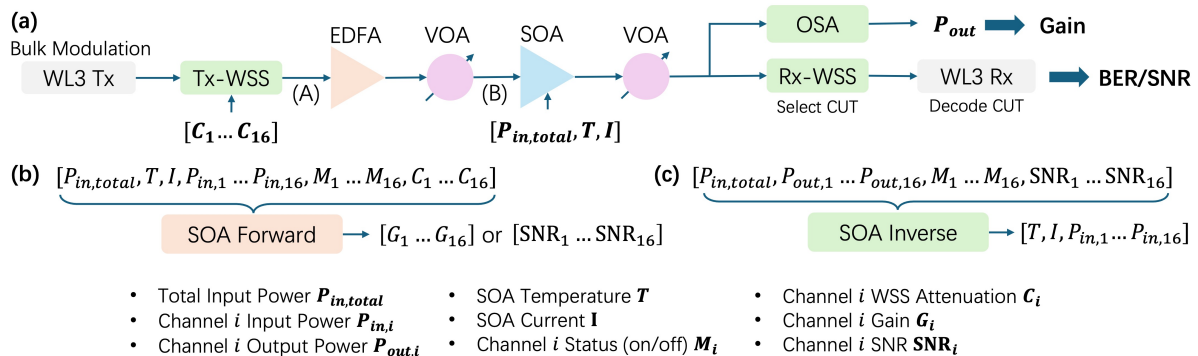


Fig. 1: (a) Schematic diagram of experimental setup for SOA WDM gain and SNR using a WaveLogic-3 Transceiver; Schematic of GPR and ANN for SOA (b) forward and (c) inverse modelling. VOA: variable optical attenuator; CUT: channel under test.

with back-to-back measurements using bulk modulation, where channel correlations are strongest [11]. The modulated signal is then passed through a wavelength-selective switch (WSS), which allows us to generate partial channel loadings through add and drop actions in addition to individual channel attenuation. The 16 WDM channels are centred around 1550 nm with a centre frequency of 193.4 THz and 50 GHz channel spacing. Each channel is modulated with a 34.5 GBaud 16QAM signal before being amplified by the SOA under test. Two distinct SOAs (Thorlabs S7FC1013) were used separately to form the dataset. Firstly, we utilise a high-resolution optical spectrum analyser (OSA, Finisar 1500S) to extract the individual channel powers before and after amplification to calculate per-channel gains. Secondly, we measure the BER at the receiver by first using a WSS for channel selection before signal demodulation with the Ciena WaveLogic 3 line card receiver. We then obtain the BER, which we convert to SNR using the relation $\text{BER} = 3/8 \times \text{erfc}(\sqrt{\text{SNR}/10})$ with erfc denoting the complementary error function.

To ensure our dataset considers a dynamic set of channel loadings and configurations, we measure both full and partial channel loadings in addition to a wide range of SOA parameter settings. For each measurement, each WDM channel has a $1/4$ probability of being turned off, with each active channel being set to an attenuation value C_i uniformly sampled from $[0, 4]$ dB in 1 dB steps. The total input power $P_{in,total}$ is uniformly sampled between $[-12, 6]$ dBm in 1 dBm steps. For each SOA, to obtain different gain characteristics, the operating conditions are varied by uniformly sampling the temperature T between $[21, 29]$ °C in 2 °C increments and the injection current I between $[150, 500]$ mA in 50 mA increments. The measured dataset consists of 800 points, equally split into 400 training and 400 testing points.

Modelling Methodology

Utilising the measured dataset, we consider a 16-channel WDM system where the SOA gain and SNR of each channel are estimated from an input condition vector as shown in Fig. 1(b). We further demonstrate inverse SOA modelling by estimating the required input channel powers $P_{in,i}$ given the target SNR_i and output powers $P_{out,i}$ for $i \in [1, 16]$, as shown in Fig. 1(c). In this paper, we demonstrate the use of GPR [12] to capture the underlying SOA effects for both the forward and inverse models. Then, ANN models are further used to benchmark the modelling performance. The GPR models used in this work consist of a constant mean function and a hybrid additive kernel function (Eq. 1) containing both the radial basis function (RBF) and the Laplacian Kernel [13] with scaling parameters σ_{rbf}^2 , σ_{lp}^2 , and length-scale ℓ_{rbf} , ℓ_{lp} . $\mathbf{x}_{i/j}$ denotes the input condition vector.

$$k(\mathbf{x}_i, \mathbf{x}_j) = \sigma_{rbf}^2 \exp\left(-\frac{(\mathbf{x}_i - \mathbf{x}_j)^2}{2\ell_{rbf}^2}\right) + \sigma_{lp}^2 \exp\left(-\frac{|\mathbf{x}_i - \mathbf{x}_j|}{\ell_{lp}}\right) \quad (1)$$

Meanwhile, the ANN counterpart model architectures are optimised by applying an open-source hyperparameter searching tool *Optuna* [14] for achieving the best performances. The loss metric used is a modified mean squared error (MSE) function, where off-channel values are excluded from the aggregated loss value. Both models are optimised using *Adam*. The GPR models are trained for 2000 iterations with a learning rate of 0.1.

Results & Discussion

First, we evaluate the forward GPR and ANN models for gain and SNR predictions in Fig. 2. In Fig. 2(a) and (e), the mean absolute error (MAE) at different numbers of training points is shown. Both figures highlight the data efficiency of the

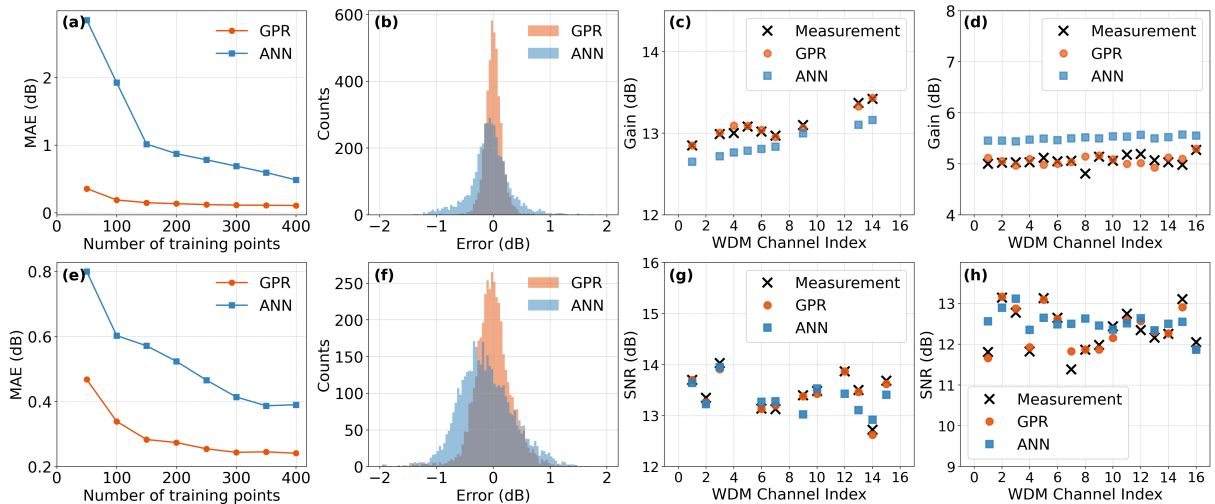


Fig. 2: Gain and SNR modelling result for GPR and ANN. (a,e) Performance vs. number of training data points exposed; (b,f) Histogram of testing dataset error; (c,g) Predictions of partially-loaded configuration; (d,h) Predictions of fully-loaded configuration.

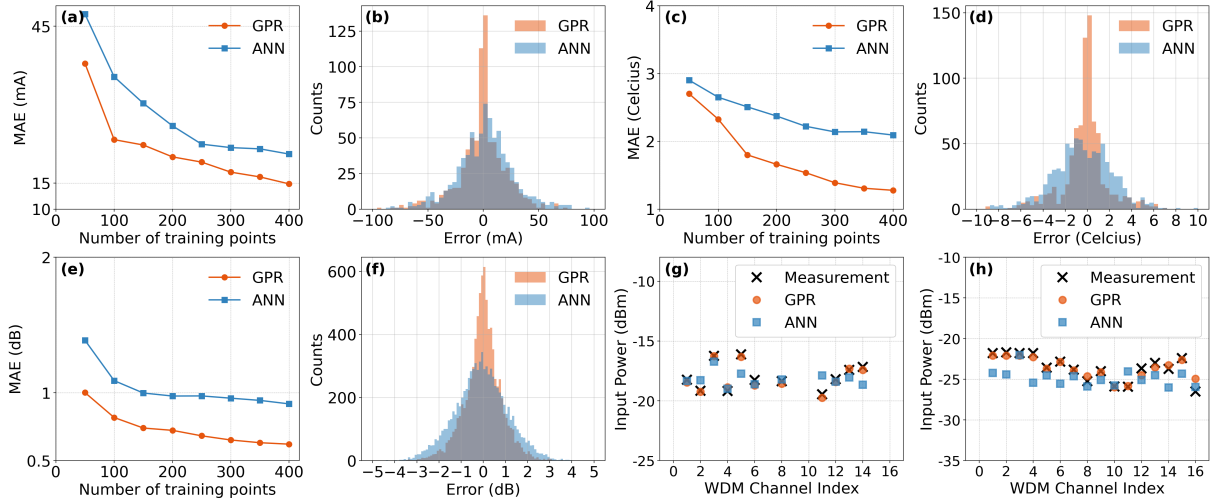


Fig. 3: Inverse modelling results for GPR and ANN models. (a,c,e) Injection current I , temperature T , and input power $P_{in,i}$ estimation performance vs. number of training data points exposed; (b,d,f) Histogram of testing dataset error; (g) Input channel power predictions of partially-loaded configuration; (h) Input channel power predictions of fully-loaded configuration.

GPR model compared with the ANN, with the ANN model typically requiring $4\times$ as much data to achieve similar accuracy. For gain estimation, the GPR model reported an MAE of 0.1 dB at 100 training points, whereas the ANN provided an MAE of 0.51 dB with all 400 training points. A further difference is seen for the SNR model, where the GPR requires only 300 training points for reaching an MAE of 0.23 dB, whereas the ANN needs 400 to reach an MAE of 0.4 dB. The histograms for the gain and SNR model prediction errors are shown in Fig. 2(b) and Fig. 2(f), respectively. While the maximum absolute gain error for the ANN model is 1.9 dB, the GPR model offers a 1.2 dB reduction to only 0.7 dB. However, for SNR estimation, both models reported a maximum absolute error of 1.9 dB. To further investigate the modelling capabilities for different configurations, we demonstrate the results for partial channel loadings in Fig. 2(c) and (g), and full channel loadings in Fig. 2(d) and (h) for gain and SNR estimation models. The gain predictions highlight a systematic underestimation by the ANN model, while the SNR predictions show a fundamental improvement from the GPR model.

The inverse modelling results are shown in Fig. 3, where we illustrate the performance of estimating the required SOA injection current, temperature, and per-channel input power for achieving a given SNR spectrum. Fig. 3 (a,c,e) shows the MAE against the number of training points (injection current, temperature, and input channel powers), highlighting both the data efficiency and improved accuracy of the GPR model. Particularly, the GPR models achieve MAE values of 17 mA, 0.9 °C, and 0.6 dB for configuration estimation. Similarly, the histograms of the testing dataset are shown in Fig. 3 (b,d,f), and the partially-loaded as well as fully-loaded input power predictions are

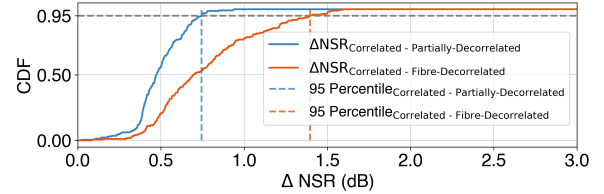


Fig. 4: CDF of ΔNSR for different decorrelation methods.

similarly shown in Fig. 3(g) and (h), respectively.

To further verify our worst-case WDM assumption and strong channel correlation effects, we experimentally compare the noise-signal ratio (NSR) differences of both partially-decorrelated and fibre-decorrelated channels. This was achieved by removing the centre bulk-modulated channel using the Tx-WSS and inserting an independent modulated centre channel at (A) in Fig. 1(a) with a 3-dB coupler. Fibre-decorrelation was then further demonstrated by inserting 20 km fibre at (B). Fig. 4 shows the cumulative density function (CDF) of the NSR differences for both decorrelation methods over 600 different channel loading configurations, further validating our worst-case assumption.

Conclusions

We experimentally demonstrate a data-driven framework for modelling SOA gain and SNR in a 16 fully/partially-loaded WDM channel system. Gaussian process regression is employed and benchmarked against optimised artificial neural network models under worst-case WDM conditions. We first demonstrate forward SOA modelling, achieving a mean absolute error of 0.1 dB for gain and 0.23 dB for SNR. We then demonstrate an inverse model to accurately estimate the required injection current, operating temperature, and input channel powers to meet specified SNR targets. This inverse model resulted in an MAE of 17 mA for injection current, 0.9 °C for temperature, and 0.6 dB for input channel power estimation.

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